

Top Insights from Chat

Hybrid RAG • Re-ranking • SQL RAG • Real World Questions

1. Chunking and Metadata Design

The largest cluster of chat questions centered on the live coding decisions around chunking structured versus unstructured sources, and how to enrich unstructured chunks with metadata.

Asked by	Question	Answer or takeaway
Aditi Reddy	Who decides what chunking strategy to use?	The developer, the AI engineer on the project, decides based on the shape of the actual data in front of them, not a fixed rule.
Anupam Saraswat	Where did we define the chunking strategy in code? Is it a default one?	No default was used. The session wrote a custom for loop over the CSV rows, since each question and answer row was treated as one self sufficient chunk.
bergelin kamdem	Why wasn't overlap used for the FAQ chunks?	Each row already stands alone as a complete question and answer pair, so there is no benefit to overlapping content between rows.
Ravindra Devagiri, Anupam Saraswat, and several others	How would you add a category field as metadata for the PDF chunks, which are not structured like the CSV?	Use the hierarchical chunk's headings and subheadings as metadata, or pass each chunk through a lightweight classification model or a small LLM call at ingestion time to assign a category. Aditya Bharadwaj's "hierarchy as metadata" answer was specifically called out as strong.
supreet kaur	Is it not overhead to use an LLM just to figure out categories?	No, because this happens once during ingestion, not on every search. Search happens far more often than ingestion, so a small model is enough and keeps retrieval fast.
Arulmurugan M	How do you fix the maximum chunk size at 128 tokens in the hybrid chunker?	It is a parameter passed to the hybrid chunker. If a hierarchical chunk exceeds that token limit, it gets split further while keeping the same section and metadata.
Tony Onabanjo	What is the limitation on the size of this in memory? Can it use streaming?	Not directly addressed live; a good follow up question for anyone working with very large document sets.

Key Insight (from Dhaval Patel)

"LangChain ultimately needs data in the form of a Document object. If the data is structured, like a CSV or JSON, we can create Document objects ourselves. If the data is unstructured, like a PDF or a wiki page, use a tool like Docling to generate those Document objects." This summary was confirmed as correct understanding during the session.

2. Dense and Sparse Embeddings, Hybrid Retrieval

Asked by	Question	Answer or takeaway
Anupam Saraswat	Can you explain the difference between tokenization and chunking again?	Tokenization breaks text into units a model can process. Chunking breaks a whole document into meaningful pieces for retrieval. Chunking happens first, then each chunk is tokenized.
AMEY BARBADIKAR	We are using sparse and dense vectorization. Where exactly is each one used during retrieval?	Both embeddings are stored for every chunk, and hybrid retrieval mode searches with both at once, then combines the results rather than picking one or the other per query.
Sharath S	Is it necessary to have both sparse and dense embeddings indexed in the same source (database)?	In this session's setup, yes, both live together in the same Qdrant collection, which is what makes the combined hybrid search possible.
Jagadeesh, Kimberly Dsouza	How do you know which chunks were retrieved through sparse embeddings versus dense embeddings?	Kimberly answered in chat: the source field in the retrieved document tells you. In this session's data, source equal to FAQ came from the CSV (sparse heavy) side and source equal to Telecom Guide came from the PDF (dense heavy) side.
Goutham	What do we do when managed services like OpenSearch are too expensive for large databases?	Not directly answered live; a good question to bring to Discord for a more specific, use case level discussion.
Gourav	Do we always have to follow these exact steps if I have a PDF and a CSV and want to use hybrid RAG?	The overall pattern (ingest, chunk appropriately per source type, embed with both a dense and sparse model, index into a hybrid enabled store) generalizes, but the specific chunking choice always depends on the shape of each source.

Standout Participant Insight (from Amit Patil)

"We are calling it hybrid RAG because we are using both dense and sparse embeddings." A simple, precise way of stating the core idea that the instructor reacted to positively in the moment.

3. Re-ranking Deep Dive

Re-ranking generated the single largest volume of chat questions in the session. The table below groups the recurring threads rather than listing every duplicate.

Asked by	Question	Answer or takeaway
Bharath Kumar	Isn't a re-ranker and the initial retriever functionally the same thing, since both use embeddings?	No. Retrieval searches a whole corpus using stored embeddings. Re-ranking works on a document already in hand, using a different model, usually a cross-encoder, to score just that document against the query.
Syed Mizbahuddin	Can we use the same embedding model for re-ranking, or does the cross-encoder need to be trained differently?	A cross-encoder is a different kind of model. It is trained to take a concatenated query and document and output one relevance score, not to produce a standalone embedding.
saikrishna	If the embedding model gives a low score to a relevant document, and it never makes the initial top K, won't re-ranking still be wrong regardless?	The instructor confirmed this is a real limitation, but said it is rare in practice because retrieval and re-ranking use two different models, so the chance of both missing the same relevant document is low.
amit s	Will re-ranking make the operation slow?	Yes, re-ranking adds latency and cost since it scores every candidate document against the query individually. This is the core trade off to weigh.
Pankaj Chand, Ankit Garg	How do you decide whether to use re-ranking at all, and what about the token and cost impact?	Base the decision on evaluation metrics such as retrieval accuracy: if accuracy is already high in a low stakes scenario, skip it; if accuracy is low, add a re-ranker. This ties directly into the upcoming evals class.
Yasmin Iqbal	Can we re-rank the documents once a day instead of on every query, to improve user experience?	No. Re-ranking has to happen per query, because it re-ranks the specific chunks retrieved for that specific question, not a static document set.
saikrishna	Doesn't embedding already read the entire document, so re-ranking means reading it twice, adding cost and latency?	Confirmed as a real trade off. This is exactly why the decision to re-rank should be evaluation driven rather than automatic.
Anurag Pandey	Is it common in industry to re-rank keyword search results and semantic search results separately?	The more common pattern is to fetch a pool from both methods, deduplicate any documents both surfaced, and re-rank the combined pool together rather than ranking each source separately.

Standout Participant Insight (from Amit Kumar)

"We use a different model for re-ranking because relying on the same model for both retrieval and re-ranking is like asking the same doctor for a second opinion. It may simply reinforce the same judgment rather than provide an independent perspective." The instructor called this a good analogy in the moment.

4. SQL RAG and MindSQL

Asked by	Question	Answer or takeaway
Kriti K	If I ask a question and the LLM gives a wrong answer, can it be told to correct itself and refer to a specific table?	Yes, in principle, by adding a memory layer to the pipeline so the model remembers the earlier question and response and can be redirected with additional context.
Pankaj Chand	SQL RAG feels very similar to keyword RAG. Can you differentiate them with a use case?	They are not similar. Keyword RAG searches text chunks for keyword matches. SQL RAG generates and executes a structured SQL query against a real database and returns a computed result.
Kaviarasu RC	What security considerations apply when using an open source GitHub library to build a Text2SQL pipeline that connects to a production database?	Do not connect an open source library directly to a production database if security is a concern. Build the query and validation logic from scratch instead.
AMEY BARBADIKAR, Sivanantham	Do we need guardrails against destructive queries like DELETE or TRUNCATE, and how would that be enforced?	Yes. Options include an intent classifier to reject malicious sounding requests early, and rule based checks such as regex to confirm a generated query has no DELETE, DROP, or similar statements before it runs.
Ankit Garg	Will the MindSQL style approach work with OracleDB or Azure SQL, given the library mentions PostgreSQL, MySQL, and SQLite?	Not directly confirmed live for those specific databases; worth testing or asking in Discord for the specific dialect support needed.
Renganathan M	Why do we need to give the LLM documentation and business context. Isn't it already trained on SQL generation?	An LLM knows SQL syntax generally, but it does not know your specific schema, table meanings, enums, or business rules unless that context is provided, the same way a new intern needs onboarding on your specific database.
Renganathan M	Do we need to tell the LLM the type of database (the dialect) when asking it to generate SQL?	Yes. Dialect matters because SQL syntax differs across MySQL, Postgres, and Microsoft SQL Server, and specifying it avoids syntax mismatches.
Sivanantham, Anurag Pandey, Yasmin Iqbal	Would this approach work for an enterprise database with many tables and schemas, complex joins, and ERD diagrams as input?	The MindSQL walkthrough was specifically shown as the answer to this class of question: index DDL, example question and SQL pairs, and documentation per table, then retrieve only the relevant subset per query rather than passing the whole schema. Whether ERD images can be used as input depends on

Asked by	Question	Answer or takeaway
		whether the model in use is a visual language model capable of interpreting them.

Standout Participant Insight (from Raghavendra)

"This is a very powerful pipeline, as it eliminates the role of data analysts who write a lot of queries, instead the system just writes them on its own from any user text." A sharp observation about the practical impact of a working Text2SQL system.

5. A Real World Scenario Worth Revisiting

From Bharath Kumar

"If I want to build a churn chat bot for a telecom company, where senior executives ask things like what is the churn this month, why is churn high, and what are the drivers of high churn, and the underlying data is customer disconnect records stored as rows and columns in a database rather than PDFs or long text, do I use RAG here, and if so, what type of RAG and what chunking strategy?" This detailed, well framed question was not directly answered live in the session. Given that the data described is structured and lives in database tables, it is a strong candidate for SQL RAG rather than a document chunking based approach, and worth raising again once the cohort has gone deeper into SQL RAG and evals.

6. Tools and Practices the Cohort Uses

Contributor	Shared context
Srinivas Vadla	Uses Snowflake Cortex at work.
Liji Alex	Mentioned Snowflake, and separately shared that their company already does similar Text2SQL style work without embeddings, still gets decent results, and ingests hierarchical markdown files that the LLM indexes and refers to directly.
Bharath Kumar	Noted that NotebookLM can generate complex SQL from plain English once given the database and table creation SQL structure.
Venkateswara Rao K	Shared that this session helped them realize a mistake made when enabling text to SQL at a previous company.